

Week 10 Applied Class Worksheet Solutions

COMP2048

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Semester 1, 2026

Task 1.

For each of the following statements, indicate whether it is **True** or **False**, and justify your answer.

- (a) If $\text{SAT} \leq_p A$, then every language in **NP** is polynomial-time reducible to A .
- (b) If one discovers a polynomial-time algorithm for any language in **NP**, then $\mathbf{P} = \mathbf{NP}$.
- (c) There exists a known **NP**-complete language that is in **P**.
- (d) It is currently unknown whether $\text{PATH} \in \mathbf{NP}$.
- (e) It is currently unknown whether every regular language has a polynomial-time verifier.
- (f) If $A \leq_p \text{PATH}$ and $A \in \mathbf{NP}$, then A is **NP**-complete.
- (g) If $\text{PATH} \leq_p B$ and $B \in \mathbf{NP}$, then B is **NP**-complete.
- (h) If **HAMPATH** can be decided in polynomial time, then $\mathbf{P} = \mathbf{NP}$.
- (i) $\text{SAT} \leq_p \text{HAMPATH}$ and $\text{HAMPATH} \leq_p \text{SAT}$.
- (j) $\text{PATH} \leq_p \text{HAMPATH}$ and $\text{HAMPATH} \leq_p \text{PATH}$.
- (k) If one proves that deciding **CLIQUE** in polynomial time is impossible, then $\mathbf{P} \neq \mathbf{NP}$.

Solution.

- (a) **True.** Since **SAT** is **NP**-complete, for every language $L \in \mathbf{NP}$ we have

$$L \leq_p \text{SAT}.$$

If $\text{SAT} \leq_p A$, then by transitivity of polynomial-time reductions,

$$L \leq_p A.$$

- (b) **False.** Discovering a polynomial-time algorithm for some language in **NP** only shows

that this particular language is in $\mathbf{P}$. To prove $\mathbf{P} = \mathbf{NP}$, one would need a polynomial-time algorithm for an $\mathbf{NP}$ -complete language.

- (c) **False.** No $\mathbf{NP}$ -complete language is currently known to be in $\mathbf{P}$. If such a language were found, then it would imply $\mathbf{P} = \mathbf{NP}$.
- (d) **False.** PATH is decidable in polynomial time. Hence

$$\text{PATH} \in \mathbf{P} \subseteq \mathbf{NP}.$$

- (e) **False.** Every regular language is decidable in linear time. Therefore every regular language is in $\mathbf{P}$, and hence in $\mathbf{NP}$. So every regular language has a polynomial-time verifier.
- (f) **False.** If $A \leq_p \text{PATH}$ and $\text{PATH} \in \mathbf{P}$, then $A \in \mathbf{P}$. This does not imply that A is $\mathbf{NP}$ -complete. To prove A is $\mathbf{NP}$ -complete, we would need every language in $\mathbf{NP}$ to reduce to A .
- (g) **False.** The fact that $\text{PATH} \leq_p B$ only shows that B is at least as hard as PATH. Since PATH is already in $\mathbf{P}$, this does not imply that B is $\mathbf{NP}$ -hard or $\mathbf{NP}$ -complete.
- (h) **True.** HAMPATH is $\mathbf{NP}$ -complete. If HAMPATH could be decided in polynomial time, then every language in $\mathbf{NP}$ would also be decidable in polynomial time. Therefore $\mathbf{P} = \mathbf{NP}$.
- (i) **True.** Both SAT and HAMPATH are $\mathbf{NP}$ -complete. Therefore

$$\text{SAT} \leq_p \text{HAMPATH}$$

and

$$\text{HAMPATH} \leq_p \text{SAT}.$$

- (j) **False.** Since $\text{PATH} \in \mathbf{P} \subseteq \mathbf{NP}$ and HAMPATH is $\mathbf{NP}$ -complete, $\text{PATH} \leq_p \text{HAMPATH}$. However, $\text{HAMPATH} \leq_p \text{PATH}$ would imply that $\text{HAMPATH} \in \mathbf{P}$, since $\text{PATH} \in \mathbf{P}$. This would imply $\mathbf{P} = \mathbf{NP}$, which is not currently known. Thus the full statement is false (or really unknown).
- (k) **True.** CLIQUE is $\mathbf{NP}$ -complete. If one proves that CLIQUE cannot be decided in polynomial time, then at least one language in $\mathbf{NP}$ is not in $\mathbf{P}$. Therefore

$$\mathbf{P} \neq \mathbf{NP}.$$